

Shubham Singh

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PROFESSIONAL SUMMARY

DevOps Engineer with hands-on experience in AWS infrastructure (EKS, EC2, VPC, Lambda), CI/CD pipeline design (Jenkins, ArgoCD, Helm), and IaC (Terraform). Built and debugged production-grade pipelines end-to-end including Jenkins Shared Libraries, GitOps with ArgoCD, and CloudWatch incident detection. Government research intern at CSIR-IIP (Govt. of India); MCA graduate, Garden City University (June 2026); immediately available.

TECHNICAL SKILLS

Cloud (AWS)	EC2, EKS, Lambda, S3, VPC, RDS, DynamoDB, IAM, CloudWatch, SNS, ALB, Route 53, EventBridge, Athena, ECR, CloudTrail
CI/CD / GitOps	Jenkins, Groovy Shared Libraries, GitHub Actions, ArgoCD, Helm, Packer, SonarQube, Trivy
Containers / IaC	Docker, Kubernetes (EKS), kubectl, eksctl, Terraform, HCL
Monitoring	CloudWatch Alarms, VPC Flow Logs, Prometheus, Grafana, Container Insights, SNS
Networking	VPC, Subnets, Route Tables, NAT/IGW, Security Groups, NACLs, ALB/NLB, DNS, TCP/IP
Scripting	Linux, Bash (set -euo pipefail), Python, sed, awk, grep, YAML, Log Analysis
Security / DB	IAM Least Privilege, Trivy (image scan), SonarQube (SAST) — MySQL, PostgreSQL, MongoDB, Redis

EXPERIENCE

Technical Intern – Automation & ML Research	Dec 2025 – Mar 2026
<i>CSIR-IIP, Council of Scientific & Industrial Research, Govt. of India</i>	Dehradun, India
- Built Battery RUL prediction pipeline on NASA PCoE dataset; evaluated 4 ML models via leave-one-battery-out CV — XGBoost champion (RMSE 23 cycles on unseen batteries). Presented at ICIEM-2026. - Automated ETL pipelines in Python/Bash (set -euo pipefail), reducing manual data handling by 40%; managed version-controlled research environments with Git.	

PROJECTS

wanderlust-infra – AWS Infrastructure & Incident Troubleshooting	<i>Terraform, EKS, CloudWatch, Jenkins, SNS</i> [github]
- Provisioned 41 AWS resources via modular Terraform (VPC, EKS, ALB, RDS, S3/DynamoDB state locking, IAM) in ap-south-1; custom Packer AMI with pre-baked Jenkins dependencies. - Diagnosed Terraform destroy failures from orphaned ALBs; built automated cleanup in Jenkins Shared Library, reducing failures to zero. Resolved EBS mount race condition causing Jenkins crashes via systemd ordering. - Configured CloudWatch alarms (EKS CPU, ALB 5xx, pod restarts) → SNS for sub-60-second incident detection.	
Wanderlust-Mega-Project – CI/CD Pipeline & GitOps Operations	<i>Jenkins, Docker, ArgoCD, Helm, SonarQube, Trivy, Prometheus</i>
- Designed 11-stage Jenkins pipeline (lint → SAST → build → Trivy scan → ECR push → GitOps update → ArgoCD sync → Prometheus/Grafana) with reusable Groovy Shared Library functions. - Diagnosed CI/CD race condition between GitOps push and ArgoCD sync; refactored into separate CI and CD Jenkinsfiles, eliminating deployment conflicts. Resolved ArgoCD OutOfSync from immutable StorageClass fields via sync-option annotation. - Integrated SNS alerts into Jenkins post-build hooks (sub-60-second notification); enforced SonarQube quality gates and Trivy image scanning as pipeline gates.	
AI-Startup-Incubator-Simulator – DevOps CI/CD Pipeline on AWS EKS	<i>Terraform, EKS, Jenkins, ArgoCD, Next.js, FastAPI</i>
- Designed EKS infrastructure via Terraform (VPC, node groups, IAM IRSA, ALB Ingress Controller) for Next.js + FastAPI + Redis + Supabase; authored full Kubernetes manifests (Deployments, Services, Ingress, HPA, Fluentd). - Established multi-repo GitOps structure (app, shared-lib, gitops, infra); managed third-party secrets (Stripe, Gemini, Anthropic) via Kubernetes Secrets.	
stock-price-pipeline – Serverless Data Pipeline on AWS	<i>Terraform, Lambda, S3, DynamoDB, Athena, EventBridge</i>
Built event-driven serverless pipeline — EventBridge → Lambda → S3/DynamoDB → Athena; converted data to Parquet with S3 Intelligent-Tiering, cutting Athena costs by 90%; Terraform-provisioned with IAM least-privilege and zero wildcard permissions.	

EDUCATION

Garden City University, Bangalore	Jul 2024 – Jun 2026
MCA (Computer Science)	Graduated June 2026
Asansol Engineering College, West Bengal	Jul 2021 – Jun 2024
BCA	GPA: 7.6/10

CERTIFICATIONS & PRESENTATIONS

- AWS Certified Cloud Practitioner (CLF-C02) – Appeared July 2026; result awaited
- Comparative Analysis of ML Models for Battery RUL Estimation* – Presented at ICIEM-2026, Garden City University, Apr 2026